

CHEMISTRY BOARD QUESTION PAPER: MARCH 2022**Time: 3 Hrs Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

- (1) Section A: Q.1 contains ten MCQs of one mark each. Q.2 contains eight very short answer questions of one mark each.
- (2) Section B: Q.3 to Q.14 are twelve short-answer questions of two marks each. (Attempt any eight.)
- (3) Section C: Q.15 to Q.26 are twelve short-answer questions of three marks each. (Attempt any eight.)
- (4) Section D: Q.27 to Q.31 are five long-answer questions of four marks each. (Attempt any three.)
- (5) Use of log table is allowed; calculator not allowed.
- (6) Figures to the right indicate full marks.
- (7) MCQs must be answered with correct option + alphabet. Only first attempt will be evaluated.

SECTION – A**Q.1. Select and write the correct answer: [10]**

(i) Coordination number in BCC:

(a)4 (b)6 (c)8 (d)12

(ii) Osmotic pressure concentration unit:

(a)Molarity (b)Molality (c)Mole fraction (d)% mass

(iii) Enthalpy change $\text{H}_2\text{O(s)} \rightarrow \text{H}_2\text{O(l)}$:

(a)Vapourisation (b)Fusion (c)Combustion (d)Sublimation

(iv) Transition element with max oxidation state:

(a)Sc (b)Fe (c)Mn (d)V

(v) Formula of sodium hexacyanoferrate(III):

(a) $\text{Na}[\text{Fe}(\text{CN})_6]$ (b) $\text{Na}_2[\text{Fe}(\text{CN})_6]$ (c) $\text{Na}_3[\text{Fe}(\text{CN})_6]$ (d) $\text{Na}_4[\text{Fe}(\text{CN})_6]$

(vi) Isopropylbenzene $\rightarrow$ (air oxidation + dilute acid):

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(a) C_6H_5OH (b) $C_6H_5COOCH_3$ (c) C_6H_5COOH (d) C_6H_5CHO

(vii) Metal nanoparticle used as water disinfectant:

(a) Carbon black (b) Silver (c) Gold (d) Copper

(viii) Acid anhydride + primary amine gives:

(a) Amide (b) Nitrile (c) Secondary amine (d) Imine

(ix) Cell potential: $Cd + Cu^{2+} \rightarrow Cd^{2+} + Cu$ ($E^\circ_{Cd} = -0.403V$, $E^\circ_{Cu} = 0.334V$)

(a) $-0.737V$ (b) $0.737V$ (c) $-0.069V$ (d) $0.069V$

(x) $[H_3O^+]$ of $0.001M$ CH_3COOH ($K_a = 1.8 \times 10^{-5}$):

(a) 1.34×10^{-1} (b) 1.34×10^{-2} (c) 1.34×10^{-3} (d) 1.34×10^{-4}

Q.2. Answer the following: [8]

(i) Product of alkyl halide + $AgNO_2$.

(ii) Name of product formed when acetone reacts with 2,4-DNP.

(iii) Name of biodegradable polyamide copolymer.

(iv) Molecularity of $NO + O_3 \rightarrow NO_2 + O$.

(v) Action of selenium on magnesium.

(vi) Isomerism in $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$.

(vii) Alloy used in Fischer-Tropsch process.

(viii) Solubility of $CH_3Br(g)$ using Henry constant ($K_H = 0.159$; pressure = 0.164 bar).

SECTION – B

(Attempt any Eight) [16]

Q.3. Explain pseudo-first order reaction with example.

Q.4. Consequences of Schottky defect with reasons.

Q.5. Action of: (i) Na/dry ether (ii) Mg/dry ether on ethyl bromide.

Q.6. Formation of peptide linkage with example.

Q.7. Derive molar mass expression using osmotic pressure.

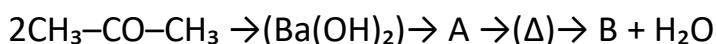
Q.8. Monodentate & ambidentate ligands with example.

Q.9. Trends in group-16: (i) Atomic radii (ii) Ionisation enthalpy (iii) Electronegativity (iv) Electron gain enthalpy.

Q.10. Preparation of phenol from aniline.

Q.11. Preparation of ethanamine from: (i) Acetonitrile (ii) Nitroethane.

Q.12. Identify A and B:



Q.13. Work done when 1 mol ideal gas expands isothermally 10L→15L at 300K.

Q.14. Moles of electrons for reduction of 2 mol Zn^{2+} ; Faradays required.

SECTION – C

(Attempt any Eight) [24]

Q.15. Chemical composition of haematite; names & e.c. of first two elements of group 17.

Q.16. Classification of polymers by structure.

Q.17. Define green chemistry; two disadvantages of nanotechnology.

Q.18. Commercial preparation of glucose; structure of adipic acid.

Q.19. Action of: (i) Hot HI (ii) PCl_5 (iii) Dil. H_2SO_4 on methoxyethane.

Q.20. Explain cationic, anionic & neutral sphere complexes with example.

Q.21. Calculate spin-only magnetic moment of M^{2+} with $Z=25$; reason why Ti^{4+} salts are colourless.

Q.22. Lanthanoid contraction; preparation of acetic acid from (i) Dry ice (ii) Acetyl chloride.

Q.23. Classification of aliphatic ketones with example; action of sodium hypoiodite on acetone.

Q.24. Define half-life of first-order reaction; derive $t_{1/2}$ and k .

Q.25. ΔH_f of $\text{CH}_3\text{OH}(\text{l})$ using given data.

Q.26. Calculate pH of buffer of 0.01M BOH & 0.02M BA ($K_b=1.8\times 10^{-5}$).

SECTION – D

(Attempt any Three) [12]

Q.27. Define isotonic solution & osmosis; calculate density of gold (FCC; $a=4.08\times 10^{-8}$ cm; $M=197$ g/mol).

Q.28. First-law equations for (i) Isothermal (ii) Adiabatic; derive relation between pH & pOH.

Q.29. Define reference electrode; functions of salt bridge; labelled diagram of SHE.

Q.30. Explain metal-deficiency defect; preparation of SO_2 ; uses of sulphur.

Q.31. Convert: (i) Ethyl bromide→ethyl methyl ether (ii) Ethyl bromide→ethene (iii) Bromobenzene→toluene (iv) Chlorobenzene→biphenyl.

